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CAPITAL ACCUMULATION AND GROWTH: A NEW LOOK AT THE EMPIRICAL EVIDENCE

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SUMMARY

Using annual data for 75 countries in the period 1960–2000, we present evidence of a positive relationship between investment as a share of gross domestic product (GDP) and the long-run growth rate of GDP per worker. This result is robust for our full sample and for the subsample of non-OECD countries, but not for the subsample of OECD countries. Our analysis controls for time-invariant country-specific heterogeneity in growth rates, and for a range of time-varying control variables. We also address endogeneity issues, and allow for heterogeneity across countries in model parameters and for cross-section dependence. Copyright © 2010 John Wiley & Sons, Ltd.

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1. INTRODUCTION

The nature of the relationship between investment in physical capital and long-run economic growth has been at the forefront of debates about development policy and has played an important role in evaluating alternative growth theories. Early endogenous growth models, notably the AK model, predicted a positive association between investment as a share of gross domestic product (GDP) and per capita GDP growth, with an increase in investment resulting in a permanently higher growth rate. Simple exogenous growth theories, notably the Solow model, predict a positive association between investment and the level of per capita GDP, but no association between investment and steady-state growth rates, with both being determined exogenously by unrelated factors. Models with exogenous growth and endogenous saving typically predict a negative association between steady-state saving and growth rates.[1]

Empirical studies suggesting that there is little or no association between investment rates and long-run growth rates appear to have been influential in these debates. Important examples include the papers by Jones (1995), Blomström *et al.* (1996), and Attanasio *et al.* (2000), who find no econometric evidence in favour of a positive and significant effect of current or past investment on steady-state growth.

Reviewing this literature, Easterly and Levine (2001) concluded that 'the data do not provide strong support for the contention that factor accumulation ignites faster growth in output per

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[1] See, for example, Carroll and Weil (1994).

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worker'.[2] If correct, this 'stylized fact' would be consistent with exogenous growth models, or perhaps with endogenous growth theories which emphasize factors other than investment in physical capital in the determination of long-run growth rates. More generally, it would question the importance of physical capital accumulation as a source of differences in long-run economic growth.

This paper reassesses the empirical basis for this conclusion. Using annual data for 75 countries in the period 1960–2000, we find strong evidence of a positive relationship between investment as a share of GDP and the long-run growth rate of GDP per worker, both for our full sample and for the subsample of non-OECD countries. For OECD countries, the evidence is weaker and not robust to alternative specifications.

Our results differ from those of previous studies partly because we do not focus only on the OECD countries, and partly because we allow for some econometric issues that were neglected in the earlier literature. We find these results when allowing for cross-country heterogeneity in all parameters, using the mean group approach of Pesaran and Smith (1995), and allowing for cross-country dependence in the shocks to the growth process, using the multifactor error structure proposed by Pesaran (2006), as well as using simpler pooled models in which slope coefficients are imposed to be common across countries. Our analysis controls for all time-invariant country-specific factors that could result in both high levels of investment and high rates of growth, and for a range of time-varying control variables that could potentially account for the relationship between investment and long-run growth rates that we find in our basic specification. We also allow for the endogeneity of current investment rates and the presence of a serially correlated error component in our econometric models of growth rates.

We do not conclude that only investment in physical capital matters. Indeed, we stress the importance of heterogeneity across countries, which may well reflect differences in economic policies and institutions. We recognize that investment may be only a proximate cause of growth, with measured variation in investment rates acting as a proxy for unmeasured variation in more primitive determinants of both investment and growth. Our results suggest that investment is at least an informative proxy and are consistent with investment being an important channel through which deeper causes influence growth outcomes. Growth theories which predict a positive relationship between investment and long-run growth rates should not be dismissed as being grossly inconsistent with the empirical evidence.

Section 2 provides a brief review of previous related research, highlighting some restrictive features of earlier specifications that will be relaxed in the models we estimate. Section 3 outlines our specifications, which allow investment to affect both the level and the growth rate of output per worker in the long run, while also allowing for business cycle dynamics. Section 4 describes our dataset and the time series properties of the main variables we use. Section 5 presents the econometric results, while Section 6 concludes.

2. RELATED LITERATURE

An important branch of the recent empirical literature on economic growth estimates specifications based on variants of the (augmented) Solow model, in which the long-run growth rate of output per worker is determined by technical progress, which is taken to be exogenous. The standard model

[2] The role of capital accumulation is similarly downplayed in King and Levine (1994) and Easterly (2001).

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used to evaluate this framework and to study the issue of (beta) convergence is derived from the transition dynamics to the steady-state growth path, as suggested by Mankiw *et al.* (1992). These models relate growth to investment, but condition on the initial level of output per worker. As a result, consistent with the underlying Solow framework, they do not allow investment to influence the steady-state growth rate.

A typical specification has the form

$$\Delta y_{it} = (\alpha - 1)y_{i,t-1} + \beta x_{it} + \gamma t + \eta_i + \varepsilon_{it} \tag{1}$$

where y_{it} denotes the logarithm of output per worker in country i at time t, Δy_{it} is the growth rate of output per worker between time $t-1$ and t, and x_{it} denotes the logarithm of the share of investment in output. Additional explanatory variables related to population growth, human capital or other factors may be included, but they do not change the essence of the points we make here. The time trend allows for a common rate of steady-state growth, and the country-specific intercept (or 'fixed effect', η_i) allows for variation across countries in initial conditions, or other unobserved factors that affect the level of the country's steady-state growth path. The residual (ε_{it}) reflects the influence of shocks that affect the (log) level of output per worker.

Cross-section studies generally focus on average growth rates measured over long periods of time, and relate these to average investment shares measured over the same period. Panel studies use repeated observations over shorter time periods, commonly 5-year averages. In this context, panel data are required to allow for unobserved heterogeneity across countries in steady-state income levels. The inclusion of time dummies rather than a simple linear trend allows for a more general evolution of total factor productivity (TFP), but still restricts TFP growth to be common across countries and independent of investment.

Equation (1) can be rewritten as

$$y_{it} = \alpha y_{i,t-1} + \beta x_{it} + \gamma t + \eta_i + \varepsilon_{it} \tag{2}$$

which is a stable dynamic model for the *level* of y_{it}, provided $|\alpha| < 1$. If we consider a steady state in which there are no shocks, the share of investment takes the constant value $x_{it} = x_i$ and output per worker grows at the common rate g, so that $y_{it} = y_{i,t-1} + g$, we obtain

$$y_{it} = \left(\frac{\beta}{1-\alpha}\right) x_i + \left(\frac{\gamma}{1-\alpha}\right) t + \frac{\eta_i - \alpha g}{1-\alpha} \tag{3}$$

This confirms that the steady-state growth rate implied by this model, $g = \gamma/(1-\alpha)$, is indeed common to all countries, and does not depend on the level of investment. For $\beta > 0$ and $|\alpha| < 1$, a permanent increase in investment predicts a higher level of output per worker along the new steady-state growth path, but affects growth rates only during the transition to the new steady state.

Both cross-section and panel studies have reported evidence that the coefficient β on measures of investment in this type of specification is positive and significantly different from zero. Examples of the former include Mankiw *et al.* (1992), Levine and Renelt (1992), DeLong and Summers (1991, 1993) and Barro and Sala-i-Martin (1995);[3] examples of the latter include Islam (1995),

[3] See also Beaudry *et al.* (2005), who suggest that the effect of investment has become larger in more recent periods.

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Caselli *et al.* (1996) and Bond *et al.* (2001). This suggests that investment affects the level of output per worker in the steady state, but does not address the question of whether investment affects the growth rate of output per worker in the long run. This is not surprising, given that these specifications are derived from the (augmented) Solow growth model in which the steady-state growth rate is taken to be exogenous.

A different branch of the empirical growth literature has focused on testing the (extended) AK model, and on Granger causality between investment and growth rates.[4] If we take first differences of equation (2) and, crucially, introduce a lagged level of investment term as an additional explanatory variable, we obtain

$$\Delta y_{it} = \alpha \Delta y_{i,t-1} + \beta \Delta x_{it} + \theta x_{i,t-1} + \gamma + \Delta \varepsilon_{it} \tag{4}$$

This is indeed a dynamic model for the *growth rate* of y_{it}, and the coefficient θ on the lagged level of investment tests whether a higher level of investment predicts a faster growth rate in the long run. If we now consider a steady state in which the share of investment takes the constant value $x_{it} = x_i$ and output per worker grows at the country-specific rate g_i, we obtain

$$g_i = \frac{\gamma}{1-\alpha} + \left(\frac{\theta}{1-\alpha}\right) x_i$$

This confirms that the steady-state growth rates implied by this model are heterogeneous and depend positively on the share of investment in output, if $\theta > 0$. The coefficient β on the change in investment again indicates whether investment affects the (log) level of output per worker along the steady-state growth path. If we replace Δx_{it} with $\Delta x_{i,t-1}$, equation (4) provides the basis for a Granger causality type test.

Both time series and panel studies have typically reported estimates of θ that are insignificantly different from zero in this type of model, suggesting that investment does not Granger-cause growth. For example, Jones (1995) finds no effect of investment on the long-run growth rate, either using annual data for individual OECD countries in the post-war period or pooling these models across countries. Similarly, Blomström *et al.* (1996), using pooled data for non-overlapping 5-year periods and a large sample of developed and less developed countries, find that investment does not Granger-cause growth, while growth does Granger-cause investment. Attanasio *et al.* (2000) find that investment Granger-causes growth, but with a negative sign.[5] While there are some exceptions in the more recent literature, such as Li (2002),[6] this econometric support for the view that higher investment does not predict faster growth appears to have been influential.

[4] The long-run relationship between saving rates and growth rates in AK-type models relies on broadening the definition of capital to include human capital, or allowing for learning-by-doing (Romer, 1986). Both extensions counteract the tendency towards diminishing returns to physical capital in the aggregate in Solow-type models.

[5] Other papers that consider Granger causality have focused on saving rather than investment. Carroll and Weil (1994) present panel data evidence for non-overlapping 5-year periods, both for OECD countries and for a wider sample, that Granger causality runs from growth to saving but not vice versa. If anything, the saving rate is negatively related to future growth. Attanasio *et al.* (2000) find significant and negative Granger causality running from saving to growth. A negative association between saving and growth is consistent with a Ramsey exogenous growth model with an infinitely lived optimizing consumer with time-separable preferences, given plausible values for the degree of risk aversion (Carroll and Weil, 1994).

[6] Related work by Kocherlakota and Yi (1997) tests whether policy variables (tax rates and public sector capital) have a significant effect on long-run growth rates. They find evidence in favour of endogenous growth for the US and the UK.

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Our derivation of this dynamic model for the growth rate suggests a potentially important problem with the least squares estimation procedures that have typically been used to obtain these results. Suppose that the shocks ε_{it} to the (log) level of output per worker, introduced in equation (1), are serially uncorrelated. The error term $\Delta\varepsilon_{it}$ in the growth equation (4) will then be serially correlated. Moreover, this model includes the lagged growth rate $\Delta y_{i,t-1}$ as an explanatory variable. This lagged dependent variable will then necessarily be correlated with the lagged shock $\varepsilon_{i,t-1}$ that appears in $\Delta\varepsilon_{it}$, rendering ordinary least squares estimates (or within estimates) of the parameters of interest biased and inconsistent. The bias in the least squares estimates of α will typically be downwards, so this approach will underestimate the degree of persistence in growth rates.[7] The bias in the estimates of β and θ is harder to characterize a priori but may be sufficiently serious to warrant further investigation.[8] In specifications where current investment is included, as in (4), consistency of least squares estimates would further require that current investment is uncorrelated with $\Delta\varepsilon_{it}$. The approach that we develop in this paper allows the long-run effects of investment on the level and growth rate of income to be estimated consistently under weaker assumptions.

A recent paper by Bernanke and Gurkaynak (2001) studies the cross-country correlation between average investment shares and average growth rates in output per worker or TFP, calculated over long periods of time. In contrast to the conclusion from the Granger causality literature, they report a significant positive correlation in both cases. While their approach avoids the estimation problem noted above, there are of course reasons to be cautious about inferring any causation from a cross-section correlation. In particular, there may be unobserved country-specific factors, such as economic, political and legal institutions, that favour both high investment and fast growth. This could account for the positive cross-section correlation even in the absence of any causal link running from investment to growth. Moreover, the use of data on actual growth rather than steady-state growth in output per worker introduces a measurement error which may be correlated with average investment rates, if periods of high (low) growth during the transition to steady-state growth paths are associated with periods of high (low) investment.

The approach we develop in the next section can be motivated either as a way of obtaining consistent estimates of the long-run 'growth effect' $\theta/(1-\alpha)$ and the 'level effect' $\beta/(1-\alpha)$ in dynamic growth models of the kind illustrated in equation (4); or alternatively as an extension of the test of the Solow model proposed by Bernanke and Gurkaynak (2001) to a dynamic panel data context, which allows us to control for unobserved country-specific factors that affect both investment and growth, as well as to allow for the likely endogeneity of current investment measures.[9] Our instrumental variable approach allows for correlation between the error term in the growth equation and all the explanatory variables (dated t or $t-1$). We further allow for unobserved heterogeneity in growth rates (i.e., γ becomes γ_i in (4)) and for heterogeneity in all the slope parameters (i.e., α, β, θ become α_i, β_i, θ_i).

[7] Both Attanasio *et al.* (2000) and Easterly and Levine (2001) emphasize the apparent lack of persistence in growth rates. Similar biases in the estimates of persistence parameters would be present if the shocks ε_{it} contain any stationary component.

[8] It is also possible that the estimate of the long-run effect of investment on growth rates ($\theta/(1-\alpha)$) could be subject to negligible bias, if the moving average component in the error term ($\Delta\varepsilon_{it}$) is well approximated by a low-order autoregressive process, and the estimated model includes sufficiently rich dynamics in $\Delta y_{i,t-k}$ and $\Delta x_{i,t-k}$ to allow for this. We thank a referee for this point.

[9] That is, the likely correlation between the current shock ε_{it} and the current investment share x_{it} in models like (1) or (4).

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3. MODEL SPECIFICATION

Our empirical analysis relies on time series variation in the data to identify the relationship between investment as a share of GDP and the growth rate of GDP per worker, while controlling for time-invariant country-specific characteristics that may affect both investment and growth. This approach allows for historical (e.g., colonial origin) and geographical (e.g., coastal access) influences on growth rates, without requiring explicit measurement of these factors. This also accounts for institutional characteristics (e.g., electoral systems, protection of property rights), to the extent that these remained constant during our sample period. This robustness to unobserved 'fixed effects' in growth rates is one of the main contributions of the paper.

Our econometric analysis uses annual data for both series. This approach is feasible in this context, since we observe significant variation over time in both annual growth rates and in annual investment shares.[10] The use of annual data provides us with enough time series observations to estimate separate models for individual countries, and thereby to allow for heterogeneity across countries in all parameters.[11] One disadvantage is that much of the annual variation in growth rates may reflect business cycle fluctuations or adjustments to occasional shocks, such as political instability or financial crises, rather than underlying long-term growth. We rely on dynamic econometric specifications to implicitly filter out these higher-frequency influences, while acknowledging the limitations of this approach. Another concern, particularly with annual data for developing countries, is that there may be considerable measurement error, although it is not clear how this would affect our results without knowing much more about the form such measurement error takes.[12]

We first outline our baseline specification for pooled annual data. Let y_{it} denote the logarithm of GDP per worker in country i in year t, and x_{it} denote the logarithm of the investment to GDP ratio. Our baseline model assumes that the behaviour of y_{it} can be represented by the ADL(p, p) (autoregressive-distributed lag) model

$$y_{it} = c_{it} + \alpha_1 y_{i,t-1} + \alpha_2 y_{i,t-2} + \ldots + \alpha_p y_{i,t-p} + \beta_0 x_{it} + \beta_1 x_{i,t-1} + \ldots + \beta_p x_{i,t-p} + \eta_i + \varepsilon_{it} \tag{5}$$

where η_i allows for time-invariant unobserved heterogeneity in income levels, and ε_{it} is a mean zero, serially uncorrelated shock which we initially assume to be independent across countries, conditional on c_{it}. Here c_{it} is a non-stationary process that determines the behaviour of the growth rate of y_{it} in the long run. The stationary ADL dynamics around this long-run income path can be interpreted in at least two ways. One possibility is to assume that economies are normally close to their long-run growth paths, except for high-frequency business cycle fluctuations.[13] In this case, the short-run dynamics are included to control for these business cycle influences in the annual data. Alternatively, we can allow some economies to be far from their long-run growth paths

[10] This approach would not be feasible if we were considering determinants of growth rates that are constant over time (e.g., legal origin) or measured infrequently (e.g., human capital), and may not be appropriate if we were considering factors that change only slowly (e.g., inequality) or infrequently (e.g., political systems). See Durlauf *et al.* (2005) for further discussion.

[11] This assumes that model parameters are stable over time, which may be a strong assumption, as discussed further below. The merits of using annual data and allowing for parameter heterogeneity are debated in Islam (1998) and Lee *et al.* (1998) in the context of estimating the speed of convergence.

[12] In an earlier version of this paper we reported broadly similar results using data averaged over non-overlapping 5- year periods, which provides a different approach to extracting long-run information, and may reduce the impact of (transient) measurement errors. Interested readers are referred to Bond *et al.* (2007) for details.

[13] Bernard and Durlauf (1996) emphasize this assumption in their discussion of time series tests for convergence.

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during our sample period, in which case the short-run dynamics must also characterize transitional growth experiences as countries adjust towards their long-run growth paths. For example, our model nests simpler dynamic specifications like equation (1) or (2), which are consistent with transitional growth in the Solow model (Mankiw *et al.*, 1992).[14] We allow for richer dynamics in our empirical models based on annual data to control also for business cycle factors.

Our baseline specification further assumes that the long-run growth rate can be modelled as

$$\Delta c_{it} = \theta_0 + \theta_1 x_{it} + d_i + e_t + v_{it} \tag{6}$$

where d_i allows for time-invariant unobserved heterogeneity in growth rates. The mean zero innovations e_t and v_{it} reflect 'permanent' shocks to the (log) level of output per worker that are respectively common to all countries (e_t) and specific to country i (v_{it}). Initially we also assume the idiosyncratic innovations (v_{it}) to be independent across countries, conditional on e_t.[15]

Our main interest centres on the parameter θ_1 in (6). In this framework, $\theta_1 > 0$ implies that a permanent increase in investment as a share of GDP predicts a higher long-run growth rate, while $\theta_1 = 0$ implies no long-run relationship. Under the null hypothesis that $\theta_1 = 0$, long-run growth rates may be different across countries (reflecting d_i) but do not vary systematically over time. Under the alternative, long-run growth rates vary systematically with investment shares—or, perhaps more plausibly, with the underlying determinants of long-run changes in investment shares. Insofar as we require the short-run dynamics in model (5) to account for transitional growth, this raises a question about whether those parameters will be stable over time.[16] A cautious interpretation of our empirical strategy would either emphasize testing the null hypothesis that $\theta_1 = 0$, or rely on the assumption that economies are sufficiently close to their long-run growth paths for unstable transitional dynamics not to be a concern.

We now outline how θ_1 can be estimated consistently, assuming that the model in (5) and (6) is correctly specified. Taking first-differences of equation (5) and substituting for Δc_{it} from equation (6), we obtain

$$\begin{aligned}\Delta y_{it} = {} & \theta_0 + \theta_1 x_{it} + \alpha_1 \Delta y_{i,t-1} + \ldots + \alpha_p \Delta y_{i,t-p} + \beta_0 \Delta x_{it} + \ldots + \beta_p \Delta x_{i,t-p} + d_i + e_t \\ & + v_{it} + \Delta \varepsilon_{it}\end{aligned} \tag{7}$$

which is a generalization of equation (4). Note that all the variables in this model are stationary, provided that the (log) share of investment in GDP is stationary.[17] Models of this form have been estimated by, for example, Jones (1995), Blomström *et al.* (1996), Attanasio *et al.* (2000), Li (2002) and Madsen (2002). In a steady state with $x_{it} = x_i$ and no shocks, output per worker grows at the country-specific rate

$$g_i = \frac{\theta_0 + \theta_1 x_i + d_i}{(1 - \alpha_1 - \alpha_2 - \ldots - \alpha_p)}$$

[14] In that context the c_{it} process would reflect the growth of total factor productivity.

[15] The estimation procedures and robustness checks that we consider in Section 5 allow us to relax the assumptions that: (i) y_{it} given c_{it} and lagged $y_{i,t-k}$ depends only on investment, and not on other measured explanatory variables; (ii) Δc_{it} depends only on investment, and not on other measured explanatory variables; (iii) the α, β and θ parameters in (5) and (6) are common to all countries; (iv) the ε_{it} and v_{it} shocks in (5) and (6) are conditionally independent across countries.

[16] Derivations of transitional growth paths, such as that in Mankiw *et al.* (1992), generally assume that long-run growth rates are time-invariant.

[17] The time series properties of our data will be considered in Section 4.

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The long-run effect of an increase in the investment share on the growth rate of output per worker is thus given by $\theta_1/(1-\alpha_1-\alpha_2-\ldots-\alpha_p)$. Given c_{it}, the long-run effect of an increase in the investment share on the (log) level of output per worker is given by $(\beta_0+\beta_1+\ldots+\beta_p)/(1-\alpha_1-\alpha_2-\ldots-\alpha_p)$.

Note that, if there are serially uncorrelated shocks ε_{it} that affect the (log) level of GDP per worker, as in (5), the error term in (7) has an MA(1) structure that makes it necessarily correlated with the lagged dependent variable $\Delta y_{i,t-1}$. More generally, the error term in the first-differenced specification would be serially uncorrelated only if there are no stationary shocks to the (log) level of output per worker. Otherwise least squares estimates of the parameters in (7) will be biased and inconsistent. Consistent estimates may be obtained, provided the shocks ε_{it} are serially uncorrelated, by using lagged values of endogenous variables from periods $t-2$ and earlier as instrumental variables; and/or by using as instruments current values of exogenous variables that are uncorrelated with v_{it}, ε_{it} and $\varepsilon_{i,t-1}$. A key issue in our empirical analysis will therefore be the availability of instrumental variables that are both valid and informative.

This specification in first-differences allows for the presence of idiosyncratic 'permanent' shocks (v_{it}) to the (log) level of output per worker. Equation (6) implies that the level of the non-stationary c_{it} process can be written as

$$c_{it} = c_{i0} + (\theta_0 + d_i)t + \theta_1 \sum_{s=1}^{t} x_{is} + \sum_{s=1}^{t} e_s + \sum_{s=1}^{t} v_{is} \tag{8}$$

Substituting for c_{it} in (5) yields a model for the level of y_{it} in which the error term has an $I(1)$ component $\sum_{s=1}^{t} v_{is}$ if these idiosyncratic permanent shocks to income levels are present. Equivalently, this implies that the $I(1)$ series y_{it} and $\sum_{s=1}^{t} x_{is}$ are not cointegrated. Consistent estimation of this model in levels would be straightforward only if there were no idiosyncratic permanent shocks to the level of c_{it}.[18]

Country-specific 'fixed effects' in income levels (η_i in (5)) are eliminated from (7) by the first-differencing transformation. Provided long time series are available, we can allow straightforwardly for unobserved 'fixed effects' in growth rates (d_i in (6)), by the inclusion of a set of country dummies in our first-differenced equations. Our use of instrumental variables (IV) estimators allows for endogeneity of current investment, in the sense that x_{it}, Δx_{it} and $\Delta x_{i,t-1}$ (as well as $\Delta y_{i,t-1}$) may be correlated with current shocks to growth (v_{it}) or with current and lagged shocks to income ($\Delta\varepsilon_{it}$). The robustness of our results to a range of alternative dynamic specifications, instrument sets, and candidate omitted variables will be considered.[19] Our empirical analysis will further allow for heterogeneity across countries in the slope parameters ($\theta_1, \alpha_1, \ldots, \alpha_p, \beta_0, \ldots, \beta_p$) of equation (7) and for cross-section dependence in the error term ($v_{it} + \Delta\varepsilon_{it}$). Details of these extensions will be presented in Section 5.

[18] In an earlier version of this paper we reported empirical results for variants of this model in levels, which were broadly similar to those presented here. Interested readers are referred to Bond *et al.* (2007) for details.

[19] That is, we will explore whether measured country-level variables other than investment should also be included in (5) or (6). This is particularly important given that the validity of any proposed instruments depends on the explanatory variables included in the model (Brock and Durlauf, 2001).

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4. THE DATA: SOURCES AND TIME SERIES PROPERTIES

We use data from both the Penn World Table 6.1 (PWT 6.1) and the World Bank World Development Indicators (WDI). Our two main variables, real GDP per worker and the share of total gross investment in GDP, are obtained from PWT 6.1. These national accounting variables are measured in constant-price international dollars. Our main instrumental variables are appropriately lagged values of the inflation rate (measured using the GDP deflator), government spending as a share of GDP, and the ratio of trade (exports plus imports) to GDP, which we obtain from WDI. In our robustness checks we also use data on the relative price of investment goods from PWT 6.1, terms of trade growth constructed from WDI, and the ratio of private credit to GDP from the World Bank Financial Structure dataset.

Our sample comprises 75 countries with annual data covering the period 1960–2000.[20] We included countries for which we have a minimum of 30 consecutive annual observations on each of these variables. Following Mankiw *et al.* (1992), we excluded countries for which oil production is the dominant industry. The excluded oil producers are Bahrain, Gabon, Iran, Iraq, Kuwait, Oman, Saudi Arabia, the United Arab Emirates and Venezuela. Similarly we excluded Botswana, because of the dominant role of mining (diamonds). Finally we dropped Nicaragua and Chad because recorded gross investment was negative in some years.

We now briefly discuss the time series properties of our main variables, the log of output per worker (y_{it}) and the log of the investment share (x_{it}). To eliminate common (to all countries) trending components, we subtract sample mean values for each year from the original series, and so investigate the time series properties of $\widetilde{y}_{it} = y_{it} - \overline{y}_t$ and $\widetilde{x}_{it} = x_{it} - \overline{x}_t$, where (e.g.) $\overline{y}_t = \frac{1}{N}\sum_{i=1}^{N} y_{it}$. We first run the augmented Dickey–Fuller (ADF) test for both the level and the first-difference of these variables, separately for each of the 75 countries, using the annual data and allowing for three lags.[21]

Table I(a) reports the number of countries for which the null hypothesis of a unit root is rejected for each series. The country-by-country ADF test cannot reject in almost all cases the presence of a unit root for both the log of GDP per worker ($\widetilde{y}_{it}$) and for the log of the investment share

Table I(A). Augmented Dickey–Fuller test

		Without trend	With trend
$\widetilde{y}$	5%	2	3
	10%	5	7
$\Delta\widetilde{y}$	5%	36	23
	10%	48	35
$\widetilde{x}$	5%	5	3
	10%	12	7
$\Delta\widetilde{x}$	5%	63	40
	10%	71	51

Notes: The figures indicate the number of countries for which the null hypothesis of a unit root is rejected at the corresponding marginal significance level.
The total number of countries is 75, and the sample period covers 1960–2000.
The number of lags used in the test is three.

[20] The Data Appendix provides a list of the sample countries and descriptive statistics for our main variables.
[21] These tests have also been run using different lag lengths, with similar results to those presented here.

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Table I(B). Im–Pesaran–Shin test for unit roots in heterogeneous panels

	Without trend	With trend
$\tilde{y}$	5.297	2.198
$\Delta\tilde{y}$	−12.809	−10.233
$\tilde{x}$	−2.845	−0.529
$\Delta\tilde{x}$	−18.419	−14.950

Notes: Reported figures are the W statistic in Im *et al.* (2003). Under the null hypothesis of a unit root (where the alternative is one-sided), the test statistic is asymptotically distributed as a standard normal.
The total number of countries is 71 (Central African Republic, Guyana, Mauritania and Papua New Guinea are dropped from the sample because of missing observations), and the sample period covers 1960–2000.
The number of lags used in the test is three.

($\tilde{x}_{it}$), whether or not a deterministic linear trend is included in the specification. When we apply the ADF test to the first-differences of these variables, we reject the non-stationarity of the growth rate of GDP per worker for 36 countries (at the 5% level) when we do not allow for a trend, and for 23 countries when we allow for a trend. The number of countries for which this test rejects the non-stationarity of the first-difference of the log investment share is larger (63 without the trend and 40 with a trend). These results suggest that the levels of both these variables are $I(1)$, and the income series may even be $I(2)$. However, these tests are known to have low power for distinguishing highly persistent, yet stationary processes from unit root processes. Moreover, there is the danger of misinterpreting structural breaks in trend stationary processes with the presence of a unit root.

We also consider a more powerful test for unit roots in heterogeneous panels, the $W_{t-\text{bar}}$ test proposed by Im *et al.* (2003). This statistic is based on an appropriately standardized average of the individual ADF statistics, and has a standard normal limiting distribution. Results are reported in Table I(b).[22] For the log of the investment share we reject the null hypothesis of non-stationarity if we rely on the version of the test that does not include a (country-specific) trend, but not if we consider the version with trends. We suspect that the former version is more appropriate for the investment share, which cannot grow (or fall) without bounds and so should have neither a deterministic nor a stochastic trend in the long run. Neither version of the test rejects the null of non-stationarity for the log level of GDP per worker, while both reject this null for the growth rate. These results suggest that the log level of output per worker may be an $I(1)$ variable, even when measured in the form of deviations from common year-specific means.

Finally we consider cointegration between the log of GDP per worker ($\tilde{y}_{it}$) and the cumulated sum of log investment shares ($\sum_{s=1}^{t} \tilde{x}_{is}$), on the assumption that these are both $I(1)$ variables. Table I(c) reports results for the panel and group ADF statistics of Pedroni (1995, 1999), both with and without country-specific deterministic trends, which test the null hypothesis of no cointegration.[23] Neither test suggests that these series are cointegrated, which is consistent with the presence of idiosyncratic permanent shocks to the (log) level of

[22] These results are based on a balanced sub-panel of 71 countries with data for all 41 years.
[23] The panel test uses a pooled specification of the ADF regression, while the group test allows for heterogeneous coefficients in the ADF regression. See Pedroni (1999, pp. 657–658) for details. These results are also based on the balanced sub-panel of 71 countries with data for all 41 years.

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Table I(C). Pedroni cointegration test between $\tilde{y}_{it}$ and $\sum_{s=1}^{t} \tilde{x}_{is}$

	Without trend	With trend
Panel ADF	1.935	1.353
Group ADF	1.429	−0.549

Notes: See Pedroni (1995, 1999). Under the null hypothesis of no cointegration (where the alternative is one-sided), the test statistic is asymptotically distributed as a standard normal.
The total number of countries is 71 (Central African Republic, Guyana, Mauritania and Papua New Guinea are dropped from the sample because of missing observations), and the sample period covers 1960–2000.
The number of lags used in the test is three.

GDP per worker, as discussed in the previous section.[24] We therefore focus our empirical analysis on IV estimates of first-differenced specifications like equation (7), in which all the variables are stationary and we can obtain consistent estimates without relying on cointegration.

5. ECONOMETRIC RESULTS

We first present the results obtained using pooled annual data. Subsequent sections present mean group results that allow for heterogeneous parameters across countries and for cross-section dependence.

5.1. Pooled Results

Table II presents estimates of our baseline specification, based on equation (7), in which the reported slope parameters are imposed to be common to all countries in the sample. A full set of year dummies is included to allow for common time effects. The set of instrumental variables used is common to all the IV results presented in this table. These instruments are dated $t-2$ and earlier, to allow for the suspected MA(1) error structure in these first-differenced models. Specifically we use lagged observations dated from $t-2$ to $t-6$ on the log of GDP per worker (y_{it}) and the log investment share (x_{it}), and lagged observations dated $t-2$ and $t-3$ on a set of additional instruments (inflation, trade as a share of GDP, and government spending as a share of GDP). The validity of these instruments is not rejected by the Sargan–Hansen test of over-identifying restrictions.[25] The validity of the exclusion restrictions required for our 'outside' instruments to be valid in these models is considered further below.

The first two columns report the results for a general dynamic specification with up to three lags of both variables (i.e., setting $p = 3$ in equations (5) and (7)). The second column includes a full set of country dummies, which allows for time-invariant unobserved heterogeneity in long-run growth rates. The third and fourth columns report the OLS estimates of these specifications

[24] Results for the group ADF test were sensitive to the choice of lag length.
[25] The reported IV results were estimated using two-stage least squares (2SLS) with the ivreg2 command in Stata. Two-step GMM results were very similar. The reported test of over-identifying restrictions is robust to heteroskedasticity across both countries and years. Degrees of freedom are 8 in the first two columns of Table II, and 12 in the last four columns of Table II.

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Table II. Baseline specifications: pooled annual models in first-differences

	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
	IV	IV with country dummies	OLS	OLS with country dummies	IV	IV with country dummies	IV	IV with country dummies
Δy_{it-1}	0.7802	0.6845	0.1081	0.0617	0.7298	0.6125	0.7270	0.6388
	(4.28)	(3.59)	(2.97)	(1.71)	(4.94)	(4.04)	(4.92)	(4.48)
Δy_{it-2}	−0.0669	−0.0706	0.0168	−0.0213	−0.0525	−0.0727	−0.0533	−0.0742
	(−1.44)	(−1.74)	(0.56)	(−0.72)	(−1.19)	(−2.02)	(−1.21)	(−2.03)
Δy_{it-3}	0.0244	0.0043	0.0519	0.0103				
	(0.62)	(0.11)	(1.99)	(0.39)				
x_{it}	0.0064	0.0212	0.0123	0.0092	0.0075	0.0224		
	(2.20)	(1.28)	(7.25)	(2.07)	(2.67)	(1.78)		
x_{it-1}							0.0075	0.0189
							(2.68)	(1.72)
Δx_{it}	0.0006	0.0571	−0.0317	−0.0300				
	(0.01)	(0.74)	(−3.03)	(−2.91)				
Δx_{it-1}	0.0762	0.0539	0.0279	0.0283	0.0764	0.0947	0.0804	0.0992
	(1.36)	(1.14)	(3.87)	(3.86)	(2.35)	(2.74)	(2.44)	(2.63)
Δx_{it-2}	−0.0085	−0.0099	0.0051	0.0064				
	(−0.83)	(−1.05)	(0.74)	(0.92)				
Δx_{it-3}	0.0122	0.0100	0.0134	0.0127				
	(1.18)	(1.00)	(1.71)	(1.60)				
Growth effect	0.0244	0.0556	0.0149	0.0097	0.0232	0.0488	0.0228	0.0433
	[0.0521]	[0.0420]	[0.0000]	[0.0351]	[0.0041]	[0.0072]	[0.0036]	[0.0127]
Level effect	0.3068	0.2912	0.0179	0.0182	0.2368	0.2058	0.2464	0.2277
	[0.3480]	[0.0250]	[0.4111]	[0.3549]	[0.1264]	[0.0015]	[0.1123]	[0.0019]
Test of first-order serial correlation	−3.28	−3.07	0.63	0.54	−3.89	−3.27	−3.83	−3.75
	[0.0010]	[0.0021]	[0.5300]	[0.5919]	[0.0001]	[0.0011]	[0.0001]	[0.0002]
Test of second-order serial correlation	0.64	0.86	0.93	−0.33	−0.59	−0.21	−0.67	−0.42
	[0.5221]	[0.3915]	[0.3523]	[0.7431]	[0.5530]	[0.8310]	[0.5056]	[0.6753]

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Table II. (*Continued*)

	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
	IV	IV with country dummies	OLS	OLS with country dummies	IV	IV with country dummies	IV	IV with country dummies
Kleibergen–Paap test of under-identification	0.6516	0.3319			0.0000	0.0043	0.0000	0.0059
Test of o.i.d. restrictions	0.4200	0.4741			0.4351	0.4155	0.4322	0.3973
Number of countries	75	75	75	75	75	75	75	75
Number of observations	2566	2566	2566	2566	2566	2566	2566	2566

Notes: A full set of year dummies is included to control for common time effects.
In the IV specifications x_{it} (or x_{it-1}), Δx_{it}, Δx_{it-1} and Δy_{it-1} are treated as endogenous. The instrument set consists of lags 2–6 of x_{it} and y_{it}, lags 2 and 3 of one plus the inflation rate, the log of trade (sum of exports and imports) as a percentage of GDP, and the log of government spending as a percentage of GDP.
The covariance matrix in all columns allows for heteroskedasticity and MA(1) errors.
t-ratios are in parentheses. The p-values of the significance tests of the long-run effects and the p-values of the serial correlation tests in Arellano and Bond (1991) are in square brackets.
p-values are reported for the Kleibergen–Paap test of under-identification, and the Sargan–Hansen test of over-identifying restrictions.

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for comparison. The next two columns report the IV estimates of a more parsimonious dynamic specification, excluding four terms that are found to be individually and jointly insignificant in the first two columns. The last two columns report the IV results for an alternative dynamic specification which includes the first lag of the log investment share ($x_{i,t-1}$) in place of the current level (x_{it}).

The comparison between the IV and the OLS results suggests that it is important to allow for an MA(1) error component in growth models of this type. As expected, we find evidence of significant, negative first-order autocorrelation in the residuals when we estimate these models using IV. Moreover, the sum of the coefficients on the lagged growth rate terms is much lower in the OLS estimates, consistent with the expected downward bias.[26] This suggests that the OLS results substantially underestimate the degree of persistence in growth rates.[27] Although the OLS results suggest a significant, positive effect of investment shares on long-run growth rates, whether or not we allow for unobserved heterogeneity in growth rates, these estimates may be biased both as a consequence of neglecting the serial correlation in the error term and more generally because current investment is likely to be correlated with current shocks to income levels or growth rates. For these reasons we focus on the IV results.

The IV estimates of our general dynamic models suggest significant, positive long-run effects of investment shares on both the slope and the level of long-run growth paths, although the level effect is significant only when we allow for unobserved 'fixed effects' in growth rates.[28] One potential concern with this general specification is that identification may be weak. Formally, the null hypothesis of under-identification is not rejected by the robust reduced rank test proposed by Kleibergen and Paap (2006). We address this concern by omitting terms that are individually and jointly insignificant, obtaining the more parsimonious specifications reported in the fifth and sixth columns of Table II.[29] This is sufficient for the Kleibergen–Paap test to now strongly reject the null hypothesis of under-identification. Inspection of the first-stage regressions also suggests that our instrument set has significant and independent explanatory power for each the three endogenous variables (x_{it}, $\Delta x_{i,t-1}$ and $\Delta y_{i,t-1}$) included in this specification.[30]

Focusing on the specification with country dummies, the long-run effects of investment shares on both the level and the growth rate of GDP per worker are found to be significant at the 1% level. The individual coefficients on $\Delta x_{i,t-1}$ and x_{it} are found to be significant at the 1% and 10% levels, respectively. The last two columns of Table II report similar

[26] For example, in the specifications with country dummies, the sum of the coefficients on the three lagged growth terms is 0.62 in the IV results and only 0.05 in the OLS results. A similar pattern was found when comparing the IV and OLS results for the more parsimonious specifications considered in Table II.

[27] We eschew the terminology 'speed of convergence' and focus primarily on the determinants of long-run growth rates, as both the level and the slope of the steady-state growth paths implied by these models are heterogeneous. See Islam (1998) and Lee *et al.* (1998) for related discussion.

[28] Standard errors for the long-run growth effect ($\theta_1/(1-\alpha_1-\alpha_2-\alpha_3)$) and level effect ($\beta_1/(1-\alpha_1-\alpha_2-\alpha_3)$) are computed using the delta method. All standard errors reported are robust to heteroskedasticity and MA(1) serial correlation.

[29] For example, a Wald test of the joint significance of the four terms ($\Delta y_{i,t-3}$, Δx_{it}, $\Delta x_{i,t-2}$, $\Delta x_{i,t-3}$) that we exclude from the model with country-specific effects has a p-value of 0.70.

[30] For example, x_{it} is significantly and positively related to past values of openness and significantly and negatively related to past values of government spending (in addition to its own lagged values and lagged values of y_{it}). Similarly, $\Delta y_{i,t-1}$ depends significantly and positively on past values of openness and significantly and negatively on past values of inflation (in addition to lagged values of x_{it}). Details are available on request.

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results for an alternative specification which uses the first lag of the log investment share ($x_{i,t-1}$), confirming that these results are not sensitive to the inclusion of the current investment term.[31]

These pooled estimates of our baseline specifications suggest an effect of additional investment on long-run growth that is quantitatively as well as statistically significant. For example, using the model reported in the last column of Table II, we find that a permanent increase in investment as a share of GDP from 9.1% (the first quartile of our sample distribution) to 15.1% (the sample median) is predicted to increase the annual growth rate of GDP per worker by about 2 percentage points. For comparison, this is about the same as the difference between the first quartile and the median of the sample growth rates.

Robustness Tests

We have considered the robustness of these results to both the choice of instruments and the inclusion of additional explanatory variables. Reducing the instrument set by dropping the second and third lags of any *one* of our 'outside' instruments (inflation, trade, government spending) produces very similar results to those presented in Table II.[32] Our main findings are not sensitive to the precise set of instruments used. Specifications which drop any *two* (or all three) of the outside instruments do not reject the null hypothesis of under-identification (using the Kleibergen–Paap test). This indicates that two or more of these outside instruments are required to identify at least some of the parameters in our model. If we augment the instrument set by including the first lag (or current level) of these outside instruments we then reject the validity of the over-identifying restrictions (using the Sargan–Hansen test), suggesting that there is contemporaneous correlation between these variables and the transient shocks to GDP per worker (ε_{it}). This last result is reassuring, indicating that the test of over-identifying restrictions does have power to detect invalid instruments in this context.

Table III considers the robustness of our results to the inclusion of additional time-varying explanatory variables. We focus on the dynamic specification reported in the final column of Table II, including country dummies, as our starting point.[33] Including the level of an additional variable ($z_{i,t-1}$) in this first-differenced specification extends the model for the long-run growth rate (equation (6)) to have the form

$$\Delta c_{it} = \theta_0 + \theta_1 x_{i,t-1} + \theta_2 z_{i,t-1} + d_i + e_t + v_{it} \tag{9}$$

Including the first-difference of an additional variable ($\Delta z_{i,t-1}$) extends the ADL model for the log of GDP per worker (equation (5)), which allows the level of this variable to affect the level of the long-run growth paths.

The first three columns consider further the validity of excluding the three variables we have relied on as outside instruments to estimate our baseline model. Consistent with the

[31] Formally this specification is consistent with a long-run growth process of the form

$$\Delta c_{it} = \theta_0 + \theta_1 x_{i,t-1} + d_i + e_t + v_{it}$$

in place of equation (6) above.

[32] These additional results are available on request.

[33] Similar results were found using alternative dynamic specifications, and omitting the country dummies.

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Table III. Robustness to additional variables (IV with country dummies)

	(i)	(ii)	(iii)	(iv)	(v)	(vi)
	Government Spending	Trade	Inflation	Relative price of investment	Terms of trade growth	Private credit
Δy_{it-1}	0.6360	0.6556	0.6467	0.5707	0.6163	0.4694
	(4.31)	(4.54)	(4.44)	(3.92)	(4.03)	(3.17)
Δy_{it-2}	−0.0644	−0.0797	−0.0724	−0.0737	−0.0782	−0.0465
	(−1.64)	(−2.03)	(−1.96)	(−2.01)	(−2.26)	(−0.95)
z_{it-1}	−0.0213	0.0063	0.0021	0.0097	0.1053	−0.0014
	(−1.32)	(0.36)	(0.19)	(0.64)	(0.88)	(−0.34)
Δz_{it-1}	−0.1402	0.0412	−0.0050	0.0115	−0.0070	−0.0189
	(−1.65)	(0.45)	(−0.25)	(0.26)	(−0.29)	(−0.75)
x_{it-1}	0.0232	0.0190	0.0185	0.0249	0.0150	0.0213
	(1.77)	(1.54)	(1.67)	(2.04)	(1.55)	(1.82)
Δx_{it-1}	0.0988	0.0987	0.0968	0.1063	0.0687	0.0851
	(2.33)	(2.30)	(2.56)	(3.06)	(2.18)	(2.62)
Growth effect	0.0541	0.0448	0.0434	0.0496	0.0325	0.0370
	[0.0188]	[0.0479]	[0.0176]	[0.0025]	[0.0467]	[0.0187]
Level effect	0.2307	0.2327	0.2275	0.2113	0.1487	0.1474
	[0.0093]	[0.0115]	[0.0028]	[0.0006]	[0.0310]	[0.0063]
Test of joint significance of z_{it-1} & Δz_{it-1}	0.2530	0.8990	0.7887	0.7657	0.5162	0.7554
Test of joint significance of x_{it-1} & Δx_{it-1}	0.0573	0.0362	0.0199	0.0077	0.0898	0.0307
Test of first-order serial correlation	−3.86	−3.69	−3.78	−3.39	−3.68	−3.03
	[0.0001]	[0.0002]	[0.0002]	[0.0007]	[0.0002]	[0.0025]
Test of second-order serial correlation	0.10	−0.41	−0.47	−0.29	0.34	−0.74
	[0.9235]	[0.6782]	[0.6381]	[0.7699]	[0.7331]	[0.4563]

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Table III. (*Continued*)

	(i)	(ii)	(iii)	(iv)	(v)	(vi)
	Government Spending	Trade	Inflation	Relative price of investment	Terms of trade growth	Private credit
Kleibergen–Paap test of under-identification	0.0416	0.0116	0.0047	0.0060	0.0437	0.0020
Test of o.i.d. restrictions	0.4775	0.2553	0.2713	0.3659	0.6288	0.0409
Number of countries	75	75	75	75	75	75
Number of observations	2562	2564	2564	2566	2420	2004

Notes: z_{it} denotes the additional variables.
A full set of year dummies is included to control for common time effects, and a full set of country dummies is included to control for time-invariant country-specific factors.
In all the specifications x_{it-1}, Δx_{it-1}, z_{it-1}, Δz_{it-1} and Δy_{it-1} are treated as endogenous. The instrument set consists of lags 2–6 of x_{it} and y_{it}, lags 2 and 3 of one plus the inflation rate, the log of trade (sum of exports and imports) as a percentage of GDP, the log of government spending as a percentage of GDP, and lags 2–4 of the corresponding new variable (z_{it}). For the specification in column (v), with terms of trade growth, lags 2 and 3 of the log of the relative price of investment goods are also included in the instrument set to achieve identification.
The covariance matrix in all columns allows for heteroskedasticity and MA(1) errors.
t-ratios are in parentheses. The p-values of the significance tests of the long-run effects and the p-values of the serial correlation tests in Arellano and Bond (1991) are in square brackets.
p-values are reported for the Kleibergen–Paap test of under-identification, and the Sargan–Hansen test of over-identifying restrictions.

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results of the Sargan–Hansen test of over-identifying restrictions in Table II, we find that the level ($z_{i,t-1}$) and the first-difference ($\Delta z_{i,t-1}$) of each of these variables are individually and jointly insignificant.[34] The estimated effects of investment on long-run growth rates and income levels continue to be positive and significant, at least at the 5% level, in all three columns. These results do *not* imply that government spending, trade and inflation are irrelevant for long-run growth rates. They do suggest that these variables have no *additional* effect on long-run growth rates, once we control for their influence on investment as a share of GDP.

The last three columns consider three additional variables: the log of the relative price of capital goods, terms of trade growth, and the log of the ratio of private credit to GDP. The choice of these three variables reflects the intersection of time-varying factors suggested by the previous literature on growth determinants,[35] and those for which we can obtain at least 30 consecutive annual observations for our sample of 75 countries. Again we find that the estimated coefficients on these additional variables are individually and jointly insignificant, while their inclusion does not affect our main finding of a significant, positive effect of investment on long-run growth rates.[36]

These results do not suggest that the significance of the relationship between investment shares and long-run growth rates in our full sample of countries is likely to be explained by the omission of other measured time-varying explanatory variables, nor is the result highly sensitive to the choice of instruments. We recognize that investment may be acting as a proxy for unmeasured variation in some more fundamental determinants of both investment and growth, on which we do not have reliable annual data for many countries. Examples could include institutional characteristics or tax structures. Our findings suggest that investment is at least an informative proxy and that changes in these deeper factors can affect long-run growth rates. While we have allowed for correlation between current investment and current shocks to growth, another interpretation would emphasize that investment may increase in anticipation of future growth. More precisely, since we instrument for current investment, this would require the lagged variables that we use as instruments to be set in anticipation of future growth, based on information that we do not control for in our models.[37] This seems unlikely but cannot be ruled out, in the absence of instruments that are both informative and strictly exogenous (i.e., unrelated to past, present and future shocks to the growth process).

Subsample Results

Table IV reports pooled IV estimates of our baseline models separately for subsamples of 20 OECD countries and 55 non-OECD countries.[38] The first and third columns report results for a

[34] The estimated coefficient on the lagged first-difference of the log of government spending as a share of GDP is just significant at the 10% level in the first column of Table III. However, this coefficient becomes insignificant (p-value = 0.22) when the level term is dropped, consistent with the result of the joint test.

[35] In particular, the papers by Fernandez *et al.* (2001) and Sala-i-Martin *et al.* (2004). Note that our specification controls for time-invariant factors. Lags of these additional variables are added to the instrument set used to obtain the results reported in these last three columns of Table III, as specified in the note to the table.

[36] These results do not depend on including only the year $t-1$ values of these additional variables. Other robustness tests that we have considered include using the level of the investment share in place of its log, using data on fixed investment in place of total investment, and including current or (long) lags of population growth as additional explanatory variables. In all cases the main results are broadly similar to those presented here.

[37] See Miguel *et al.* (2004) for a related discussion.

[38] The 20 high-income countries included in our OECD subsample are indicated in the Data Appendix.

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Table IV. Pooled annual models: OECD and NON-OECD subsamples

	(i)	(ii)	(iii)	(iv)
	OECD		NON-OECD	
Δy_{it-1}	0.8171	0.7892	0.6903	0.6649
	(3.79)	(5.57)	(3.54)	(4.41)
Δy_{it-2}	−0.1934	−0.1490	−0.0685	−0.0705
	(−1.93)	(−2.38)	(−1.62)	(−1.82)
Δy_{it-3}	0.0542		0.0030	
	(0.87)		(0.07)	
x_{it}	0.0273	0.0293	0.0227	
	(1.20)	(1.70)	(1.34)	
x_{it-1}				0.0197
				(1.68)
Δx_{it}	0.0850	0.0582	0.0489	
	(1.31)	(1.02)	(0.65)	
Δx_{it-1}	−0.0037		0.0657	0.1033
	(−0.06)		(1.39)	(2.62)
Δx_{it-2}	0.0191		−0.0082	
	(0.75)		(−0.83)	
Δx_{it-3}	0.0055		0.0094	
	(0.30)		(0.87)	
Growth effect	0.0848	0.0816	0.0606	0.0485
	[0.0732]	[0.0455]	[0.0348]	[0.0133]
Level effect	0.3287	0.1617	0.3084	0.2547
	[0.1013]	[0.2075]	[0.0319]	[0.0047]
Test of first-order serial correlation	−3.50	−4.20	−2.91	−3.60
	[0.0005]	[0.0000]	[0.0036]	[0.0003]
Test of second-order serial correlation	−0.97	−1.42	0.83	−0.32
	[0.3319]	[0.1558]	[0.4060]	[0.7488]
Kleibergen–Paap test of under-identification	0.0480	0.2910	0.3088	0.0109
Test of o.i.d. restrictions	0.5864	0.7457	0.5041	0.5106
Number of countries	20	20	55	55
Number of observations	697	697	1869	1869

Notes: A full set of year dummies is included to control for common time effects, and a full set of country dummies is included to control for time-invariant country-specific factors.
In all the specifications x_{it} (or x_{it-1}), Δx_{it}, Δx_{it-1} and Δy_{it-1} are treated as endogenous. See Table II for the list of instruments.
The covariance matrix in all columns allows for heteroskedasticity and MA(1) errors.
t-ratios are in parentheses. The p-values of the significance tests of the long-run effects and the p-values of the serial correlation tests in Arellano and Bond (1991) are in square brackets.
p-values are reported for the Kleibergen–Paap test of under-identification, and the Sargan–Hansen test of over-identifying restrictions.

general dynamic specification, with country dummies included, separately for these subsamples. The second and fourth columns report results for our preferred parsimonious specifications for these subsamples.

Perhaps unsurprisingly, the results for the non-OECD countries are very similar to those for the full sample, and are not discussed further. However, the evidence of a long-run effect of investment on growth rates is weaker and less robust for the OECD subsample. While this effect is significant at the 5% level in the specification we report in the second column of Table II, we note that this result is sensitive to the precise dynamic specification. The long-run growth effect becomes insignificant for the OECD subsample in specifications that omit the current level of investment

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term (x_{it}). This provides a first hint that the relationship between investment and growth may be weaker in the higher-income countries.[39]

5.2. Results for Individual Countries and Mean Group Estimates

In this subsection we re-estimate our baseline model using annual time series data for individual countries. This relaxes the restriction that the slope parameters are common across countries. If that restriction is invalid, the inferences we draw from the pooled specifications may be incorrect. For example, what we have called 'the' long-run growth effect may be an inconsistent estimate of the average effect (across countries) of investment on long-run growth rates.[40] Heterogeneity in this parameter is certainly plausible; this may, for example, reflect differences across countries in the efficiency with which investment spending is allocated, which in turn may be related to the institutional and policy environment. The differences between the results for the OECD and non-OECD subsamples in Table IV already suggest that parameter heterogeneity may be a relevant concern here.

To account for the likely importance of common shocks, we initially use the transformed variables $\widetilde{y}_{it}$ and $\widetilde{x}_{it}$ as defined in Section 4, subtracting the sample mean values for the same year from the original series. If the slope coefficients are identical across countries, this controls for common factors that affect growth in all countries in the same way, and is equivalent to the inclusion of year dummies in the pooled models we presented in Table II. If the slope coefficients differ across countries, this procedure is only an approximate way to deal with such common influences.

The dynamic specification that we consider here is

$$\Delta\widetilde{y}_{it} = \theta_{0i} + \theta_{1i}\widetilde{x}_{it} + \alpha_{1i}\Delta\widetilde{y}_{i,t-1} + \alpha_{2i}\Delta\widetilde{y}_{i,t-2} + \beta_{0i}\Delta\widetilde{x}_{it} + \beta_{1i}\Delta\widetilde{x}_{i,t-1} + \widetilde{v}_{it} + \Delta\widetilde{\varepsilon}_{it} \tag{10}$$

In contrast to the pooled results for the full sample, preliminary investigations suggested that both the current $\Delta\widetilde{x}_{it}$ and the lagged $\Delta\widetilde{x}_{i,t-1}$ terms were significant in the time series models for some countries. This model is estimated separately for each country, using 2SLS to account for the MA(1) component in the error term, with the same instrument set used previously to obtain the pooled results in Table II.

The first column of Table V presents a summary of the results. We report mean and median values across the sample of 75 countries of the main coefficients, and of the long-run growth and level effects. Due to the presence of a few outliers in the individual coefficient estimates that distort significantly the unweighted means, we report here robust estimates of the means, together with their standard errors.[41] This is a robust variant of what Pesaran and Smith (1995) call the Mean Group estimator.

These results are qualitatively similar to our pooled IV results for the full sample, in that on average we find significant persistence in growth rates, and significant effects of investment on

[39] The coefficient on the level of investment term, and the associated long-run growth parameter, are also found to be insignificant when these specifications are estimated using OLS for the OECD subsample. This finding is consistent with that reported by Jones (1995).

[40] See Pesaran and Smith (1995) for a discussion of the biases that can result from not recognizing the heterogeneity of slope coefficients in dynamic panel data models. See also Lee *et al.* (1997) for an application to the issue of convergence.

[41] The robust estimate of the mean is obtained using the *rreg* command in Stata. The *rreg* command performs a robust regression, based initially on Huber weights and then on biweights. When no explanatory variables are specified, *rreg* produces a robust estimate of the mean. For details, see Hamilton (1991).

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Table V. Mean group estimates

		(i)	(ii)	(iii)	(iv)	(v)
		Full sample	OECD (subtracting full sample means)	Non-OECD (subtracting full sample means)	OECD (subtracting OECD means)	Non-OECD (subtracting Non-OECD means)
θ_1	Mean	0.0116	0.0268	0.0106	−0.0079	0.0117
		(2.70)	(3.31)	(2.09)	(−0.70)	(2.21)
	Median	0.0092	0.0293	0.0087	−0.0056	0.0085
$\beta_0 + \beta_1$	Mean	0.0722	0.1952	0.0296	0.1206	0.0322
		(6.41)	(9.39)	(2.25)	(4.58)	(2.45)
	Median	0.0546	0.1938	0.0266	0.1389	0.0283
$\alpha_1 + \alpha_2 - 1$	Mean	−0.7319	−0.8218	−0.6972	−0.6224	−0.7117
		(−21.25)	(−12.61)	(−17.21)	(−10.46)	(−17.38)
	Median	−0.7091	−0.7535	−0.6515	−0.6421	−0.6613
Growth effect	Mean	0.0210	0.0301	0.0176	−0.0110	0.0191
		(3.40)	(3.46)	(2.27)	(−0.36)	(2.52)
	Median	0.0219	0.0449	0.0166	−0.0086	0.0156
Level effect	Mean	0.0955	0.2132	0.0458	0.1662	0.0449
		(5.24)	(7.80)	(2.02)	(3.96)	(1.97)
	Median	0.1014	0.2197	0.0361	0.1925	0.0413

Notes: The number of countries is 75, 20 and 55, respectively.
Robust means are reported. t-ratios are in parentheses.
All the specifications include $\tilde{x}_{it}$, $\Delta\tilde{x}_{it}$, $\Delta\tilde{x}_{it-1}$, $\Delta\tilde{y}_{it-1}$ and $\Delta\tilde{y}_{it-2}$, with $\tilde{x}_{it}$, $\Delta\tilde{x}_{it}$, $\Delta\tilde{x}_{it-1}$ and $\Delta\tilde{y}_{it-1}$ treated as endogenous. See Table II for the list of instruments used (here in deviations-from-means form).

both the level and the slope of long-run growth paths. However, the magnitude of these mean (or median) values of the country-specific parameters differs substantially from the pooled estimates. Allowing for parameter heterogeneity across countries suggests that growth rates are typically less persistent than suggested by the pooled IV estimates, although still rather more persistent than suggested by the pooled OLS estimates.[42] The average long-run effects of investment on income levels and growth rates are approximately half the size of the pooled IV estimates for our parsimonious specifications in Table II, but still substantial. A permanent increase in investment as a share of GDP from 9.1% (sample first quartile) to 15.1% (sample median) is thus predicted to increase the annual growth rate of GDP per worker by about 1 percentage point in the typical country, but with considerable heterogeneity across countries in the likely magnitude of this effect.

The second and third columns of Table V report the robust means and medians of these parameter estimates separately for the subsamples of 20 OECD and 55 non-OECD countries. For this specification, the average of the long-run growth effects is found to be positive and significant for both subsamples. However, this result is again more fragile for the OECD countries. If we account for common influences by subtracting the subsample means rather than the full sample means from the original series, the mean long-run growth parameter remains positive and significant for the non-OECD countries but becomes insignificantly different from zero for the OECD countries. These results are presented in the final two columns of Table V. In contrast to the pooled results in Table IV, we do find that the average of the long-run effects of investment on

[42] For example, the robust mean of the sum of the coefficients on the lagged growth rate terms is about 0.27 in the full sample results in Table V (the median is about 0.29). The pooled IV estimates in Table II are between 0.54 (column (vi)) and 0.62 (column (ii)), while the pooled OLS estimate is about 0.05 (column (iv)).

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the level of income per worker is positive and highly significant in the OECD subsample, using either of these specifications.

Cross-Section Dependence

All the specifications we have considered in the preceding sections have maintained that the time-varying shocks to the income processes are either common to all countries (i.e., e_t in (6)) or independent across countries (i.e., ε_{it} in (5) and v_{it} in (6)). Pesaran (2006) has recently suggested a simple procedure to allow for a form of cross-section dependence with a multiplicative factor structure. Analogously to allowing each country to have its own slope coefficients on the observed explanatory variables, such as investment shares or lagged growth rates, this allows each country to have its own slope coefficient on the unobserved common factor. Thus, for example, the term e_t in (6) or (7) can be replaced by a term of the form $\phi_i e_t$. This specification is less restrictive than imposing a uniform response to the common shocks, for example allowing different responses to oil price shocks for countries with differing reliance on imported oil, or different responses to global shocks depending on institutional quality.

This form of cross-country dependence can be allowed for by including as additional regressors, in the time series models for each country, the sample means for each year of the dependent variable and each of the explanatory variables. The time series $\bar{y}_t = \frac{1}{N}\sum_{i=1}^{N} y_{it}$ and $\bar{\underline{x}}_t = \frac{1}{N}\sum_{i=1}^{N} \underline{x}_{it}$ are thus used as additional regressors, where y_{it} again denotes the dependent variable and $\underline{x}_{it}$ denotes the vector of explanatory variables. This formulation can also be thought of as a flexible way of introducing trends when estimating separate models for each country. The mean group estimates are then obtained in the usual way, by taking the (robust) mean across countries of the coefficients in the augmented time series models for individual countries.

The first column of Table VI reports the full-sample results for the same dynamic specification considered in Table V, allowing for this form of cross-section dependence. Again the results are qualitatively similar to our earlier full-sample results. The robust mean of our key long-run growth effect is positive and significant, and slightly larger than the corresponding estimate in Table V.

The second and third columns of Table VI report the robust means and medians of these parameter estimates separately for the OECD and non-OECD subsamples. The average of the long-run growth parameters remains positive and significant for the non-OECD countries, but becomes insignificant for the OECD countries when we allow for cross-section dependence in this way. The final two columns of Table VI show that similar results are obtained here when we include subsample means rather than full-sample means in the time series models for the individual countries. Again we find that the average of the long-run effects of investment on the level of GDP per worker is positive and highly significant in the OECD countries.

Overall the results in this section confirm the robustness of our main result for our full sample of 75 countries and for the subsample of non-OECD countries, but indicate that the evidence for a significant effect of investment on long-run growth rates is substantially weaker for the subsample of OECD countries. This difference between OECD and non-OECD countries echoes the results of Evans (1998), based on the time series properties of the per capita income series for different countries.[43] In particular, Evans (1998) found that income differences between countries are stationary in countries with high levels of education, consistent with exogenous growth models with common long-run growth paths. However, he found that the income series are not cointegrated

[43] The analysis of Evans (1998) builds on that of Bernard and Durlauf (1995).

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Table VI. Mean group estimates, allowing for cross-sectional dependence

		(i)	(ii)	(iii)	(iv)	(v)
		Full sample	OECD (with full sample means)	Non-OECD (with full sample means)	OECD (with OECD means)	Non-OECD (with Non-OECD means)
θ_1	Mean	0.0164	0.0078	0.0198	0.0040	0.0150
		(3.07)	(0.83)	(3.04)	(0.38)	(1.95)
	Median	0.0207	0.0095	0.0250	0.0085	0.0138
$\beta_0 + \beta_1$	Mean	0.0470	0.1511	0.0064	0.1196	0.0121
		(4.24)	(7.27)	(0.49)	(5.63)	(0.91)
	Median	0.0385	0.1334	−0.0054	0.1103	0.0042
$\alpha_1 + \alpha_2 - 1$	Mean	−0.7992	−0.6788	−0.8397	−0.6569	−0.7907
		(−25.29)	(−12.89)	(−21.82)	(−11.51)	(−20.69)
	Median	−0.8091	−0.7495	−0.8202	−0.6915	−0.7603
Growth effect	Mean	0.0288	0.0109	0.0360	0.0081	0.0237
		(2.25)	(0.57)	(2.12)	(0.49)	(1.82)
	Median	0.0295	0.0217	0.0408	0.0119	0.0189
Level effect	Mean	0.0734	0.2041	0.0209	0.1804	0.0342
		(4.10)	(4.85)	(1.15)	(5.22)	(1.60)
	Median	0.0664	0.2167	−0.0037	0.1692	0.0042

Notes: The number of countries is 75, 20 and 55, respectively.
Robust means are reported. t-ratios are in parentheses.
All the specifications include x_{it}, Δx_{it}, Δx_{it-1}, Δy_{it-1} and Δy_{it-2}, as well as the sample means of these explanatory variables and the dependent variable Δy_{it}. These means are calculated either for the full sample or the respective subsamples.
x_{it}, Δx_{it}, Δx_{it-1} and Δy_{it-1} are treated as endogenous. See Table II for the list of instruments used.

in countries with low levels of education, as predicted by endogenous growth models that allow for different long-run growth paths.

Other possible explanations for this difference between our results for OECD and non-OECD countries include different roles for investment in catch-up growth and in growth at the frontier, and/or the differential importance of some factor such as capital market imperfections in generating the relationship between investment and long-run growth rates. For example, Aghion *et al.* (2006) have recently presented a model of catch-up growth in the presence of financing frictions in which domestic saving affects FDI and growth only in countries that are far enough below the frontier.[44]

Finally, we note that this difference may be related to earlier results in DeLong and Summers (1993), who found that the relationship between equipment investment and growth is stronger in less developed countries.[45] While their specifications conditioned on the initial level of income, and so did not necessarily identify effects of investment on long-run growth rates, there is nevertheless a broad similarity between their finding and our results here.

6. CONCLUSIONS

In contrast to the suggestion that capital accumulation plays only a minor role in economic growth, the main finding of this paper is that investment as a share of GDP has a large and statistically

[44] The mechanism suggested in their model is that higher domestic savings in relatively poor countries increases the number of joint projects that local banks can co-finance and monitor effectively.
[45] See also Auerbach *et al.* (1994) for a critique of this study, and DeLong and Summers (1994) for a reply.

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significant effect, not only on the level of output per worker but more importantly on long-run growth rates. This result is robust to controlling for unobserved country-specific differences in long-run growth rates, and for a range of additional time-varying control variables that could potentially account for the relationship.

These findings are consistent with the predictions of several types of endogenous growth models, such as AK-type models (Romer, 1986), and Schumpeterian-type models in which capital is used as an input in the production of innovations (Howitt and Aghion, 1998). They are more challenging to rationalize in the context of Solow or Ramsey-type models, in which long-run growth rates are considered exogenous.

We find robust evidence of this relationship using data for our full sample of 75 countries, and for the subsample of non-OECD countries. The result is more fragile for the subsample of OECD countries, echoing earlier differences between related empirical results for countries at different stages of development in, for example, Evans (1998) and DeLong and Summers (1993). This may reflect important differences in the growth process, depending on the distance of a country from the frontier.

We emphasize that this evidence does not rule out a very important role for many other factors, such as the quality of economic, legal and political institutions, and the nature of both macroeconomic and microeconomic government policies, including those related to education and research. Such factors may play a key role in determining investment levels, and may account for some of the heterogeneity across countries that we find in the magnitude of the relationship between investment shares and long-run growth rates. These are important issues for future research. These factors may also affect long-run growth rates directly, in ways captured by the country effects and time effects in our specifications. Moreover, we recognize that measured variation in the share of investment in GDP may be acting as a proxy for unmeasured variation in some of these factors. At least we have shown that investment is an informative proxy. Without convincing annual data for many countries on the quality of institutions and policies, it will remain challenging to make much stronger claims about the identification of the true causal determinants of economic growth. In the meantime, our findings challenge the influential view in some of the recent literature suggesting that variation in investment shares is unrelated to variation in long-run growth rates.

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DATA APPENDIX

Argentina	France[a]	Netherlands[a]
Australia[a]	Ghana	Niger
Austria[a]	Greece[a]	Nigeria
Bangladesh	Guatemala	Norway[a]
Belgium[a]	Guyana	Pakistan
Benin	Honduras	Papua New Guinea
Bolivia	Hong Kong	Paraguay
Brazil	Iceland[a]	Peru
Burkina Faso	India	Philippines
Burundi	Indonesia	Rwanda
Cameroon	Ireland[a]	Senegal
Canada[a]	Israel	South Africa
Central African Rep.	Italy[a]	Spain[a]
Chile	Jamaica	Sri Lanka
China	Japan[a]	Sweden[a]
Colombia	Kenya	Switzerland[a]
Congo, Republic of	Korea, Republic of	Syria
Costa Rica	Luxembourg[a]	Thailand
Cote d'Ivoire	Madagascar	Togo
Denmark[a]	Malawi	Trinidad and Tobago
Dominican Republic	Malaysia	Turkey
Ecuador	Mali	United Kingdom[a]
Egypt	Mauritania	United States of America[a]
El Salvador	Mexico	Uruguay
Finland[a]	Morocco	Zambia

[a] These countries are included in the OECD subsample.

Descriptive statistics

	Growth rate of output per worker	Investment share
Mean	0.0149	16.1222
Standard deviation	0.0533	8.6837
First quartile	−0.0082	9.0993
Second quartile	0.0186	15.0506
Third quartile	0.0420	22.5991
Min.	−0.5430	0.9453
Max.	0.2765	46.4931
No. of observations	2566	2566

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